Erlan Cassiano

Space Systems Engineer | Mechanical Architect

Satellite Lifecycle | MTP Architecture | Requirements & V&V | Budgets & ICDs | Supplier Follow-up

São José dos Campos, SP, Brazil | +55 12 99793-3231 | erlan.neo@gmail.com

linkedin.com/in/erlancassiano | erlancassiano.github.io | Nationality: Brazilian

PROFESSIONAL SUMMARY

Mechanical engineer with experience since 2014: Mechanical Architect and Space Systems Engineer at Visiona (since 2023) and propulsion trainee at Thales Alenia Space (2014–2016). Covers the lifecycle from requirements to qualification on geostationary (Brazil's SGDC) and low Earth orbit (LEO) Earth-observation (SATVHR, sub-300 kg) missions; leads SATVHR's mechanical and thermal architecture, mass budget and interfaces. At Thales, showed that more than 16 kg of additional propellant could be loaded, adding ~2.5 months of life (~R$36M).

CORE COMPETENCIES

- **Systems engineering:** requirements specification, mass budget, configuration trade-offs, interface control documents (ICDs), verification and validation (V&V) to tailored ECSS (European Cooperation for Space Standardization) standards, supplier specification and follow-up.
- **Reviews:** preliminary requirements review (PRR), preliminary design review (PDR) and critical design review (CDR); test readiness review (TRR) and test review board (TRB) for environmental tests; non-conformance reviews.
- **Subsystems:** Mechanical, Thermal and Propulsion (MTP) architecture, spacecraft structures, jitter-aware structural design, thermal control, bipropellant propulsion.
- **AIT & verification:** assembly, integration and test (AIT), mechanical ground support equipment (MGSE), structural-thermal model (STM) sine, random and sine-burst qualification to NASA GEVS (General Environmental Verification Standard) levels, jitter measurement, test data.
- **Leadership & tools:** technical leadership, team training, stakeholder coordination, Scrum; finite element analysis (FEA) with Femap/Nastran, thermal models (Thermal Desktop/SINDA), CAD (Creo Parametric, Onshape, SolidWorks), MATLAB, Python.

PROFESSIONAL EXPERIENCE

Visiona Tecnologia Espacial S.A. **Mar 2023 – Present**

Space Systems Engineer – Mechanical Architect *São José dos Campos, Brazil (hybrid)*

- Lead the mechanical and thermal architecture of SATVHR, a sub-300 kg, 5-year-life LEO satellite in phase B/C; own its mass budget, trade-offs and ICDs.
- Manage 4 junior engineers and 2 interns since Mar 2023 (trained 2); present mechanical and thermal at system reviews (PRR, PDR) towards the system, structure and thermal CDRs (end 2026).
- Wrote, with the team, the specification of the platform thermal controller built by a supplier (hardware and software); followed its design reviews, tested the unit and sent corrections through to acceptance.
- Directed and reviewed structural and thermal analyses, running and building FEM and thermal models (Femap/Nastran, Thermal Desktop/SINDA) for design reviews and troubleshooting; authored margin reports.
- Co-authored the STM test plan and specification and led its vibration qualification at INPE/LIT (Integration and Testing Laboratory), passed May 2026, correlating results with the FEM.
- Led the concept of the lifting and handling MGSE for the sub-300 kg satellite, detailed and manufactured by a subcontracted supplier; tested it with the team during the STM AIT campaign.

Visiona Tecnologia Espacial S.A. **Jan 2023 – Mar 2023**

Space Systems Engineer (VCUB1 CubeSat) *São José dos Campos, Brazil*

- VCUB1 (12 kg CubeSat, operated in orbit): supported final integration, incl. the delta vibration and solar-panel deployment tests; revised the solar-panel assembly procedure and performed the final flight assembly.
- Ran the 6U deployer-rail fit-check, reviewed qualification test data and prepared integration tooling for the Falcon 9 launch (SpaceX Transporter-7 rideshare, Apr 2023).
- Co-authored the IAC 2024 paper on the VCUB1 thermal-vacuum test campaign (IAC-24-B4.IP.21), doing the data analysis and part of the writing.

CNPq / AEB / INPE – SINDAE project **Oct 2016 – Sep 2017**

SET-G Fellow – Satellite Engineering *Brasília, Brazil*

- Held a one-year CNPq (Brazil's national research council) SET-G fellowship in satellite engineering: supported the development of educational devices and modelled missions, engines and launchers for investment planning.

Thales Alenia Space **Mar 2014 – Oct 2016**

Satellite Propulsion Engineer – SGDC Technology Transfer (on-the-job training) *Cannes, France*

- Showed that more than 16 kg of additional propellant could be loaded on SGDC, accounting for tank deformation under pressurisation, adding ~2.5 months of service life (~R$36M).
- Selected by the Brazilian Space Agency (AEB) for the SGDC Technology Absorption Programme: ~5,500 h of full-time on-the-job training with Thales Alenia Space engineers on the SGDC propulsion subsystem.
- Worked across the bipropellant propulsion lifecycle (tanks, valves, lines, interfaces), from requirements specification and design to production, procedures, testing and operation.
- Assessed propulsion performance over the satellite's life (pressure, thruster performance, temperature variation, propellant gauging) under ECSS-driven V&V.
- Witnessed, as a trainee, all propulsion-related SGDC tests and selected alignment and thermal tests during the satellite's test campaign in Cannes.

OTHER EXPERIENCE

Junior Space Project Analyst, Funcate (Brazilian space-science foundation) Feb 2020 – Nov 2020, Brasília
AEB: model-rocketry launch documentation; Alcântara Launch Centre authorisations.

RuaDois: Project Coordinator, Head of Product, COO, Technical Consultant Oct 2018 – Jul 2024, São Paulo
Team grew to 20 staff + 50+ inspectors and visit companions; 20,000+ properties and 6,000+ inspections in 1.5 years.

Co-founder & COO, CIV: equipment-rental marketplace; Scrum Master Nov 2016 – May 2018, Brasília

Co-founder & CTO, Fox17: cement-industry filters and separators Jul 2015 – Jun 2023, Brasília
Designed 80 t, 100 m+ steel chimneys and their flanges for export.

Project Designer, Autotrac: SolidWorks test-rig structures; CNC prototypes Feb 2014 – Apr 2014, Brasília

EDUCATION

Instituto Tecnológico de Aeronáutica (ITA), São José dos Campos, Brazil **Jul 2025 – Present**
M.Sc. in progress (expected Jul 2027), EAM-1 – Aeronautical Design, Structures, and Aerospace Systems
Thesis: *Development of a Magnetohydrodynamic (MHD) Magnetic-Fluid Reaction Wheel for CubeSat Attitude Control*; advisor: Prof. Tiago de Paula.

Universidade Cruzeiro do Sul, Online **Jun 2019 – Nov 2021**
MBA in Data Analysis, BI and Big Data (450 h)

Universidade de Brasília (UnB), Brasília, Brazil **Jun 2008 – Jun 2013**
Bachelor's in Mechanical Engineering
Research volunteer, hybrid rocket static-fire test benches (Mar 2012 – Jun 2013); metrology and reverse-engineering intern (Jan 2012 – Mar 2013). Extracurricular: Apuama Racing (Formula SAE), N-Prize.

PUBLICATIONS

- Silva, S. et al., incl. **Cassiano, E.** (2024). *Vcub1 Brazilian CubeSat Thermal Vacuum Tests Campaign.* 75th International Astronautical Congress (IAC 2024), Milan, IAC-24-B4.IP.21. doi:10.52202/078365-0126.
- Bertoldi, A. E. M.; **Cassiano, E. R. S.** et al. (2016). *A Study of the Atomization Process Using Numerical Simulation.* CILAMCE 2016, RIPE (UnB) v.2 n.31. CFD (ANSYS CFX) of a pressure-swirl atomiser.
- **Cassiano, E.** (2013). *Projeto de uma Bancada Experimental para o Teste de Motores Foguetes com Propulsão Híbrida* (hybrid rocket motor test bench). 19th UnB Scientific Initiation Congress.

CERTIFICATIONS & COURSES

Designing Space Missions and Systems (Teaching Science & Technology Inc., 2024, 40 h) • CNC machining-centre programming, setup and operation for aerospace (SENAI, Brazilian industrial training service, 2024, 120 h); CAM programming (SENAI) • Creo Parametric (Mult-e, 2023, 20 h) • Space Mission Structures (Applied Technology Institute, 2023, 40 h)

HONOURS & AWARDS

- AEB recognition, Mar 2017: propellant-loading method that extends the lifetime of the SGDC satellite.
- FAP-DF (Federal District Research Support Foundation) funding, Sep 2019: Omega electromechanical battery for photovoltaic systems (technical lead, Fox17).

VOLUNTEERING

COBRUF (Brazilian university rocketry competition): Launch Base Subsystem Lead, design of a mobile launch platform (Oct 2013 – Jan 2016) • N-Prize team (UnB): co-founder, systems engineering and hybrid rocket motor (Jun 2010 – Jul 2012)

LANGUAGES

Portuguese: native • English: professional working proficiency (Duolingo English Test 129, Aug 2026) • French: limited working proficiency